



Vibration of a Cantilever Plate (Level 7)

Engineering Computation (Assignment: MATLAB/Vibration)

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Abstract

This report examines the vibrational behaviour of a cantilever plate through three different approaches: an analytical model implemented in MATLAB, a numerical modal analysis carried out in SolidWorks, and a series of laboratory experiments. The natural frequencies and associated mode shapes obtained from each method were compared to evaluate their accuracy and consistency.

The experimental measurements—based on observing how sand redistributes over the plate at resonance—served as the reference for assessing the numerical predictions. Among the two computational tools, SolidWorks provided the closest correspondence with the experimental results, successfully reproducing both the frequency values and the overall shape of the vibration modes. MATLAB, while effective at estimating the lower modes quickly, showed increasing deviations for higher-order modes due to the simplifications inherent in its reduced analytical model.

Overall, the study underscores the complementary nature of theoretical, numerical and experimental methods in vibration analysis. SolidWorks proved to be the most practical and accurate tool for capturing realistic mode shapes—particularly at higher frequencies—while the experimental results remained essential for validating and interpreting the performance of the computational models.

Chapter 1

Introduction

Cantilever structures are fundamental elements in modern engineering and architecture, enabling systems to support loads without intermediate supports. Their ability to transfer bending moments and shear forces into a single fixed boundary allows for efficient and visually unobstructed designs in applications such as overhanging balconies, extended roof elements and metallic plates used in industrial machinery (Rao, 2011).

The cantilever principle is not only an engineering concept but also one found in nature. Tree branches function as natural cantilevers, resisting self-weight and wind-induced oscillations through hierarchical fibre arrangements that balance rigidity with controlled flexibility (Vincent, 2012). Similar behaviour can be observed in the exoskeletal projections of certain crustaceans, which distribute stresses through lightweight yet mechanically optimised biological materials (Gibson, 2010). These natural analogues demonstrate the effectiveness of the cantilever mechanism in both biological and engineered systems.

A clear engineering example that illustrates the relevance of cantilever behaviour is the wing of the *Boeing 787 Dreamliner*. Acting as a large composite cantilever extending from the fuselage, the wing benefits from advanced carbon-fibre materials that reduce weight while increasing flexibility. However, this configuration requires accurate vibration modelling to prevent aeroelastic instabilities such as flutter, emphasising the importance of modal analysis in high-performance structures (Thomson and Dahleh, 1997).

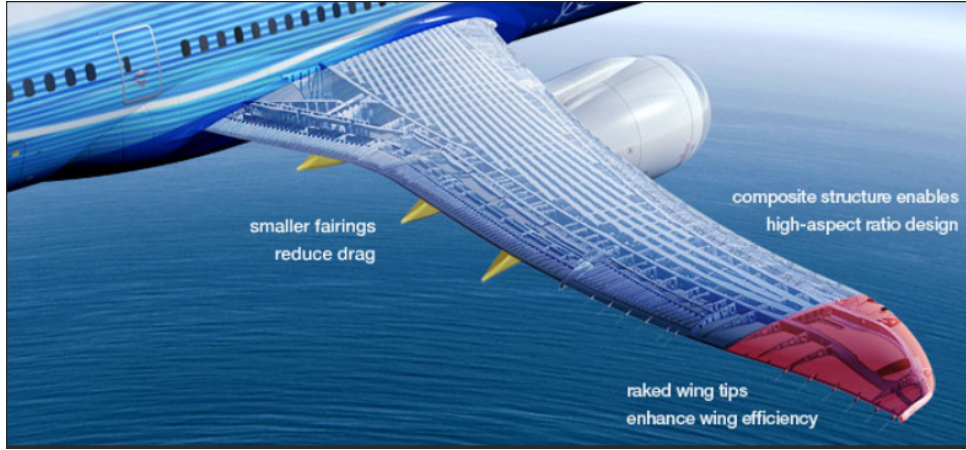


Figure 1.1: The Boeing 787 Dreamliner wing, a composite cantilever requiring precise vibration control.

These examples highlight that studying the vibrational behaviour of cantilever plates is essential for ensuring structural performance, durability and safety. A metal plate in a machine assembly, for instance, may experience fatigue under repeated vibrations if its modal characteristics are not correctly understood. Likewise, insufficient stiffness in a cantilevered architectural element may lead to long-term deformation or cracking. For these reasons, the accurate prediction of natural frequencies, mode shapes and static deflections is a key aspect of structural design.

In this context, the present report investigates the modal and static behaviour of a cantilever plate using MATLAB-based numerical models, finite element simulations in SolidWorks and experimental measurements. By comparing these approaches, the study examines how closely theoretical predictions, computational simulations and laboratory data agree when analysing a reduced-degree-of-freedom structural system. This comparison provides insight into the reliability and limitations of common analytical tools used in structural vibration analysis.

Chapter 2

Theoretical Background

A cantilever plate behaves as a multi-degree-of-freedom (MDOF) structure whose response under dynamic and static loading depends on its stiffness distribution, mass properties and boundary conditions. Understanding its behaviour requires two essential analytical approaches: modal analysis, which characterises the free vibration properties of the plate, and static analysis, which evaluates deformation under time-independent forces. These principles constitute the foundation of structural dynamics as established in classical works by Rao (7), Thomson and Dahleh (8), Cook (2), and Clough and Penzien (1).

2.1 Modal Analysis

The free vibration of an undamped n -degree-of-freedom system is governed by the matrix equation derived from Newton's second law:

$$M\ddot{x}(t) + Kx(t) = 0, \quad (2.1)$$

where M is the mass matrix, K the stiffness matrix, and $x(t)$ the displacement vector. Classical structural dynamics references (e.g. Rao (7)) show that assuming harmonic motion,

$$x(t) = \phi e^{j\omega t}, \quad (2.2)$$

leads to the substitution:

$$\ddot{x}(t) = -\omega^2 \phi e^{j\omega t}. \quad (2.3)$$

Substituting this into (2.1) yields the generalised eigenvalue problem:

$$(K - \omega^2 M)\phi = 0, \quad (2.4)$$

which admits non-trivial solutions only when:

$$\det(K - \omega^2 M) = 0. \quad (2.5)$$

Each solution ω_i corresponds to a natural frequency, and each associated vector ϕ_i is a mode shape describing a characteristic deformation pattern of the plate.

2.1.1 Normal Mode Vibration

Normal mode vibration refers to the condition in which all points in the structure oscillate synchronously while maintaining fixed amplitude ratios. Following classical vibration theory (Rao (7); Thomson & Dahleh (8)), the displacement vector for free vibration can be expressed as:

$$x(t) = XA \sin(\omega t + \phi), \quad (2.6)$$

where:

- X is the modal vector (relative amplitudes),
- A is the amplitude,
- ω is the natural frequency,
- ϕ is the phase angle.

Differentiating (2.6) twice yields:

$$\ddot{x}(t) = -\omega^2 XA \sin(\omega t + \phi) = -\omega^2 x(t). \quad (2.7)$$

Substituting (2.7) into the free-vibration equation (2.1), one obtains:

$$(K - \omega^2 M)X = 0. \quad (2.8)$$

Equation (2.8) forms the mathematical basis of modal analysis: natural frequencies arise as the roots satisfying the determinant condition, and mode shapes emerge as the corresponding eigenvectors.

2.1.2 Eigenvalue Reformulation

By multiplying (2.8) on the left by M^{-1} , the system can be written in the standard eigenvalue form:

$$(M^{-1}K - \lambda I)X = 0, \quad \lambda = \omega^2. \quad (2.9)$$

Defining the matrix:

$$A = M^{-1}K, \quad (2.10)$$

the problem becomes:

$$AX = \lambda X, \quad (2.11)$$

where the eigenvalues λ_i correspond to the squared natural frequencies, and eigenvectors X_i represent the vibrational mode shapes. This formulation underpins numerical solvers such as MATLAB's `eig(K,M)` function.

2.2 Compliance Matrix

The compliance matrix H is defined as the inverse of the stiffness matrix:

$$H = K^{-1}, \quad (2.12)$$

and relates applied forces to resulting displacements through:

$$\delta = Hf. \quad (2.13)$$

For the present work, six measurement points are considered on the cantilever plate, yielding a 6×6 experimental compliance matrix:

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{14} & h_{15} & h_{16} \\ h_{21} & h_{22} & h_{23} & h_{24} & h_{25} & h_{26} \\ h_{31} & h_{32} & h_{33} & h_{34} & h_{35} & h_{36} \\ h_{41} & h_{42} & h_{43} & h_{44} & h_{45} & h_{46} \\ h_{51} & h_{52} & h_{53} & h_{54} & h_{55} & h_{56} \\ h_{61} & h_{62} & h_{63} & h_{64} & h_{65} & h_{66} \end{bmatrix}.$$

Each displacement entry is computed as:

$$\delta_i = \sum_{j=1}^6 h_{ij} f_j. \quad (2.14)$$

2.3 Static Analysis

Static analysis evaluates deformation under constant forces. Under linear elastic behaviour, Hooke's law applies:

$$\sigma = E\varepsilon, \quad (2.15)$$

where σ is stress, ε is strain, and E is Young's modulus.

Using finite element principles (Cook (2)), the static equilibrium equation is expressed

as:

$$Ku = F, \tag{2.16}$$

where u is the displacement vector and F the load vector.

The global stiffness matrix is assembled from:

$$K = \int_V B^T D B dV, \tag{2.17}$$

with B the strain–displacement matrix, D the material constitutive matrix, and V the element volume.

This framework is implemented in MATLAB using numerical solvers, and in commercial FEM software such as SolidWorks Simulation (8).

Chapter 3

Experimental Methodology

The experimental phase of this work aimed to identify the natural frequencies and vibration mode shapes of the aluminium cantilever plate through controlled excitation and visual inspection of its nodal patterns. This hands-on approach complements the theoretical and numerical models by providing direct physical insight into the dynamic behaviour of the structure.

3.1 Materials and Equipment

Cantilever Plate

The tested component was a 306×306 mm aluminium plate with a thickness of 1.6 mm, rigidly clamped along one edge to reproduce a classical cantilever configuration with six discretised degrees of freedom, as described in the laboratory presentation (?).

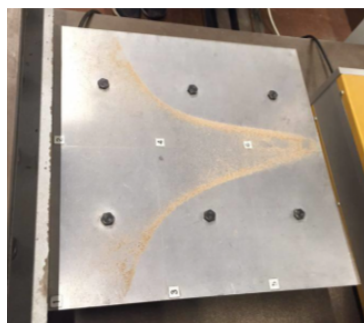


Figure 3.1: Aluminium cantilever plate clamped at one end before testing.

Electromagnetic Exciter

An electrodynamic shaker was used to induce controlled vibrations in the plate by converting the input electrical signal into harmonic mechanical motion.



Figure 3.2: Electrodynamic shaker used to excite the cantilever plate.

Function Generator

The function generator provided the excitation signal and allowed fine adjustment of the frequency to locate the resonance peaks.



Figure 3.3: Function generator used to control the excitation frequency.

Digital Calliper

A digital calliper was used to accurately measure the plate dimensions and verify the spacing between mounting holes.



Figure 3.4: Digital calliper used to measure the plate geometry.

Digital Dial Gauge

A digital dial gauge was available to measure small static displacements and to verify the sensitivity of the setup.



Figure 3.5: Digital Gauge

Digital Scale and Triple Beam Balance

A digital scale and a triple beam balance were used to determine the mass of the plate and any additional loads used in the experiment.

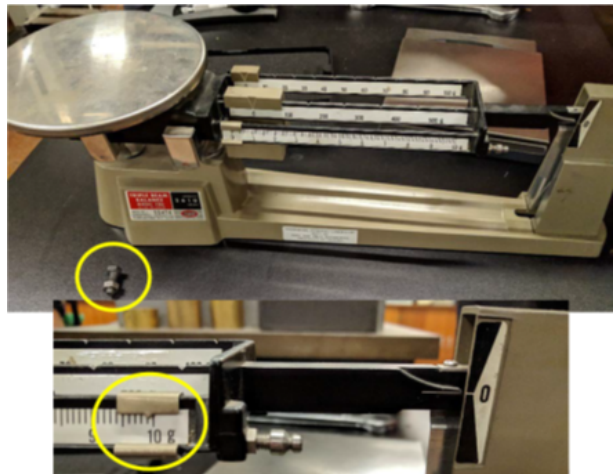


Figure 3.6: Digital scale (left) and triple beam balance (right) used for mass measurements.

3.2 Experimental Procedure

1. Plate Preparation

A thin layer of dry sand was uniformly spread across the surface of the plate. This technique is effective for visualising nodal lines, as sand accumulates naturally along regions of minimal displacement during resonance.



Figure 3.7: Cantilever plate covered with a thin layer of sand for mode shape visualisation.

2. Measurement of Physical Properties

The plate dimensions (length, width and thickness) were recorded using the digital caliper, while its mass was measured using the scale and triple beam balance. These values are required for subsequent calculation of the plate's inertial and stiffness properties.

3. Controlled Excitation

Following the laboratory guidelines (?), the electrodynamic shaker was driven from approximately 5 Hz upwards until the first nine resonance frequencies were identified. At each resonance, the excitation frequency was carefully adjusted using the function generator to stabilise the observed mode shape.

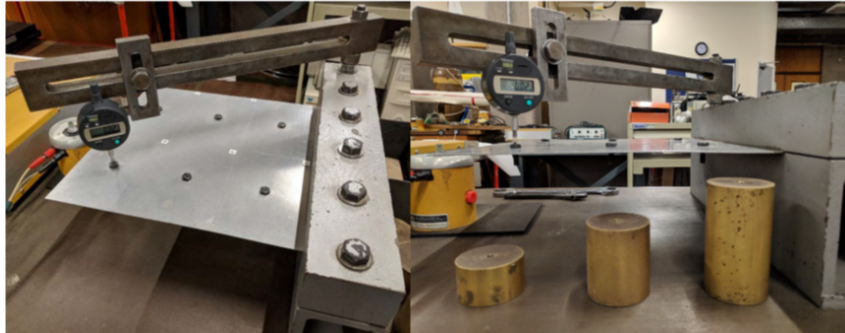


Figure 3.8: Complete experimental setup including the plate, shaker and function generator.

4. Mode Shape Visualisation

At each identified natural frequency, the plate vibrated in a distinct mode, and the sand redistributed toward the nodal lines. The resulting patterns were photographed and later compared with theoretical and numerical mode shapes obtained from the 6-DOF model and finite element simulations.

Chapter 4

Matlab Procedure

4.1 Detailed Description of the MATLAB Code

This section presents a step-by-step explanation of the MATLAB script used for the theoretical modal analysis of the cantilever plate with six degrees of freedom (6-DOF). The code directly implements the concepts introduced in the theoretical background: construction of the mass matrix, derivation of the flexibility matrix from static measurements, solution of the eigenvalue problem, and three-dimensional visualisation of the mode shapes.

4.1.1 Environment Initialisation and Output Logging

The script begins by cleaning the graphical environment and preparing a log file to store the command window output:

```
1 % Close all open figure windows
2 close all
3
4 % Delete previous output file (if it exists)
5 delete 6DF_Plate.txt
6
7 % Start logging command window output to text file
8 diary 6DF_Plate.txt
```

- `close all` closes any figures left open from previous runs, preventing confusion between old and new plots.
- `delete 6DF_Plate.txt` removes any existing output file with that name, so that the current run starts with a clean log.

- `diary 6DF_Plate.txt` activates the *diary* mode: everything printed in the command window is also written to the text file. This provides a complete record of the numerical results without manual copying.

In this way, each execution starts in a controlled environment and produces a traceable output file.

4.1.2 Physical Parameters and Mass Matrix Construction

The next part of the script defines the physical properties of the system: aluminium density, bolt mass, and the mass of the material removed to create the bolt hole. These are used to construct a consistent inertial model of the plate.

```

1 % mb = mass of bolt [kg]
2 % mbh = mass of hole for bolt [kg]
3 % mp = mass of 12"*12"*1.6mm solid aluminium plate [kg]
4 % ms_dev6 = mass of 1/6 plate + mass of bolt [kg]
5
6 % Density of aluminium [kg/m^3]
7 densAL = 2770
8
9 % Mass of one bolt [kg]
10 mb = 9.3*1e-3
11
12 % Mass of material removed for bolt hole [kg]
13 % PI * (hole radius [mm])^2 * plate thickness [mm]
14 % * density of aluminium[kg/m^3] * 1e-9
15 mbh = pi*3*3*1.6*densAL*1e-9;
```

- `densAL` stores the aluminium density in kg/m^3 , required to convert volumes into masses.
- `mb` is the mass of a single bolt in kilograms, either measured or taken from manufacturer data.
- `mbh` computes the mass of aluminium removed to drill the bolt hole:

$$V_{\text{hole}} = \pi r^2 t \quad [\text{mm}^3],$$

$$m_{\text{hole}} = V_{\text{hole}} \rho_{\text{Al}} \cdot 10^{-9} \quad [\text{kg}],$$

where the factor 10^{-9} converts mm^3 to m^3 . This allows the model to account not only for the added mass of the bolts, but also for the missing material at the hole locations.

Using these quantities, the total plate mass and the effective mass associated with each of the six degrees of freedom are computed:

```

1 % density of aluminium[kg/m^3]*width[mm]*length[mm]*thickness[mm
  ]*10^-9
2 mp = densAL*(306*306*1.6)*1e-9
3
4 % ms_dev6 = mass of 1/6 of the vibrating system
5 ms_dev6 = mp/6 + mb - mbh
6
7 % m = mass matrix
8 m = ms_dev6*eye(6)
9
10 format short e

```

- `mp` calculates the mass of a solid $306 \times 306 \times 1.6$ mm plate:

$$m_p = \rho_{Al} L W t \cdot 10^{-9},$$

where L , W and t are in millimetres and the factor 10^{-9} converts the volume into m^3 .

- `ms_dev6` represents the mass assigned to one of the six discrete segments of the vibrating system:

$$m_{1/6} = \frac{m_p}{6} + m_b - m_{\text{hole}}.$$

In other words, each station is assumed to carry one sixth of the plate mass, plus one bolt, minus the missing aluminium due to the corresponding hole.

- `m = ms_dev6*eye(6)` builds the 6×6 mass matrix M as a diagonal matrix:

$$M = m_{1/6} I_6.$$

Each degree of freedom is assigned the same lumped mass, and no off-diagonal inertial coupling terms are included. This is a classical *lumped mass* approximation.

- `format short e` switches the numerical display to scientific notation, which makes small or large values easier to read.

This diagonal mass matrix greatly simplifies the eigenvalue analysis, at the expense of some approximation with respect to a fully consistent mass matrix.

4.1.3 Applied Loads and Measured Displacements

The following block introduces the experimental data used to construct the flexibility matrix: the applied static loads and the corresponding measured displacements:

```
1 % Applied loads [kg]
2 loads = [1.482 1.482 0.976 0.976 0.976 0.976
3          1.482 1.482 0.976 0.976 0.976 0.976
4          0.976 0.976 0.976 0.976 0.976 0.976
5          0.976 0.976 0.976 0.976 0.976 0.976
6          0.976 0.976 0.976 0.976 0.498 0.498
7          0.976 0.976 0.976 0.976 0.498 0.498];
8
9 % Measured displacements [m]
10 dispt = [0.170 0.050 0.310 0.185 0.500 0.345
11          0.050 0.170 0.185 0.310 0.345 0.500
12          0.300 0.180 1.830 1.240 3.420 2.670
13          0.180 0.300 1.240 1.830 2.670 3.420
14          0.420 0.345 3.315 2.505 3.800 3.140
15          0.340 0.420 2.505 3.315 3.140 3.800] *1e-3;
```

- **loads** contains the nominal masses (in kg) applied at different loading positions for the six static test configurations. Each row corresponds to a measurement station, and each column to one load case.
- **dispt** stores the measured displacements for each station and load case. The values are initially specified in millimetres and then multiplied by $1e-3$ to convert them into metres.

It is important to note that the numerical values in the **loads** matrix may differ slightly from those reported in the assignment brief. This discrepancy arises from:

- limited resolution and calibration uncertainties of the digital scale used in the laboratory,
- small manufacturing tolerances in the masses of the standard weights,
- rounding of the measured values when entering the data into MATLAB.

For the purpose of this study, the values entered in the **loads** matrix are treated as the most accurate representation of the real applied loads, since they are directly based on the measured masses rather than on nominal values.

4.1.4 Computation of the Flexibility Matrix H

The flexibility (or compliance) of the structure is defined as the displacement produced by a unit force. Using the measured displacements and the applied loads, the script computes the experimental flexibility matrix H according to:

$$h_{ij} = \frac{\delta_{ij}}{F_{ij}}, \quad F_{ij} = 9.81 \cdot \text{loads}(i, j).$$

In the code, this is implemented as:

```
1 % h = flexibility matrix [m/N]
2 h = zeros(6,6);
3 for i = 1:6
4     for j = 1:6
5         h(i,j) = dispt(i,j)/(9.81*loads(i,j));
6     end
7 end
8 h
9
10 format
```

- `h` is initialised as a 6×6 matrix of zeros.
- For each row i (measurement point) and column j (load case):
 - `dispt(i,j)` is the measured displacement δ_{ij} in metres.
 - `9.81*loads(i,j)` is the applied force F_{ij} in newtons, using $g = 9.81 \text{ m/s}^2$.
 - Their ratio gives h_{ij} in m/N, which is stored in `h(i,j)`.
- The resulting matrix H therefore captures the static compliance of the plate at the six measurement stations under all applied load cases.
- The command `format` restores the default numerical display format in the command window.

This experimentally derived flexibility matrix is the starting point for obtaining the stiffness matrix $K = H^{-1}$ of the 6-DOF model.

4.1.5 Eigenvalue Problem and Natural Frequencies

Once the mass matrix M and flexibility matrix H are known, the modal properties of the system can be obtained by solving the associated eigenvalue problem. The script forms the dynamic matrix:

$$A = M^{-1}H^{-1} = M^{-1}K,$$

and computes its eigenvalues and eigenvectors:

```

1 % Solve eigen value problem to obtain matrix of eigen vectors
2 % (displacement amplitudes [m]) and matrix of eigen values
3 % (resonant frequencies[rad/s]
4 [v,d] = eig(inv(m)*inv(h));
5
6 % Convert rad/s to Hz and display resonant frequencies
7 Hz = sqrt(ones(1,6)*d)/(2*pi)
8
9 % Display signed displacement amplitude vectors
10 v
11
12 diary off

```

- `inv(h)` computes the inverse of the flexibility matrix, i.e. the stiffness matrix $K = H^{-1}$.
- `inv(m)*inv(h)` builds the matrix $A = M^{-1}K$, which appears in the standard eigenvalue formulation.
- `[v,d] = eig(...)` solves the eigenvalue problem

$$Av = \lambda v.$$

- `d` is a diagonal matrix whose entries λ_i are the eigenvalues.
- `v` is the matrix whose columns v_i are the eigenvectors, i.e. the mode shapes expressed as displacement amplitude ratios at the six DOFs.
- In the theoretical framework, the eigenvalues satisfy

$$\lambda_i = \omega_i^2,$$

where ω_i is the angular natural frequency in rad/s.

- The line

$$\text{Hz} = \text{sqrt}(\text{ones}(1,6)*d)/(2*\text{pi})$$

implements the conversion

$$f_i = \frac{\sqrt{\lambda_i}}{2\pi},$$

yielding the natural frequencies f_i in hertz. The vector `Hz` therefore contains the six natural frequencies of the 6-DOF model.

- Finally, the eigenvectors $\mathbf{v}$ are displayed, and `diary off` closes the log file.

Although, numerically, using `inv()` is not the most robust approach, it clearly reflects the theoretical relation $A = M^{-1}H^{-1}$ and is adequate for a system of this modest size.

4.1.6 Three-Dimensional Plotting of Mode Shapes

The last part of the script focuses on transforming the mode shape vectors into three-dimensional surfaces for visual interpretation.

```

1 % Plot 3-D plots of eigen vectors
2 scrsz = get(0,'ScreenSize');
3 x = [0 0; 2 2; 6 6; 10 10];
4 y = [-3 3; -3 3; -3 3; -3 3];
5 cl = [1 1
6       2 2
7       3 3
8       6 6];
9
10 for p = 1:6
11     r = [0 0];
12     c = v(:,p);
13     n = size(c);
14     j = 1;
15     z = zeros(n(1)/2,2);
16     for i = 1:2:n(1)
17         z(j,1) = c(i);
18         z(j,2) = c(i+1);
19         j = j+1;
20     end
21     z = [r; z];
22     pos = 50 + p*20;
23     figure('name','3-D Plot of Peak Displacements','NumberTitle',
24            'off', ...
25            'Position',[pos pos scrsz(3)/2 scrsz(4)/2]);
26     surface(x,y,z,cl)
27 % Ground plane for zero displacement reference
28 hold on
29 mesh(x,y,zeros(4,2))
30 hold off
31 txt = sprintf('%0.4f Hz',Hz(p));
32 title(txt)
33 axis([0 10 -3 3 -1 1])

```

```

33 end
34
35 clear
36 clc

```

- `scrsz = get(0,'ScreenSize')` retrieves the screen dimensions so that each figure window can be positioned neatly on the display.
- The matrices `x` and `y` define a simple rectangular grid representing the projected geometry of the plate in the longitudinal and transverse directions.
- `c1` assigns colour levels that will be used to highlight differences in displacement amplitude across the surface.
- In the loop `for p = 1:6`, each mode p is processed in turn:
 - `c = v(:,p)` extracts the eigenvector associated with mode p , i.e. the relative displacements at the six DOFs.
 - The nested loop reorganises the components of `c` into the matrix `z`, arranging the values in pairs so that they can be mapped onto the grid defined by `x` and `y`.
 - The row `r = [0 0]` is prepended to `z` to introduce a fixed reference line at the clamped end of the plate.
- `surface(x,y,z,c1)` generates a coloured 3D surface where:
 - the x -axis represents the longitudinal position along the plate,
 - the y -axis represents the transverse direction,
 - the z -axis represents the mode shape amplitude (relative displacement).
- `mesh(x,y,zeros(4,2))` draws a flat “ground plane” at $z = 0$ to visualise the zero-displacement reference configuration and to emphasise the magnitude of the deformation.
- The title of each figure is set using `sprintf` to include the corresponding natural frequency in Hz, making it easy to identify which mode is being plotted.
- `axis([0 10 -3 3 -1 1])` fixes the axis limits so that all mode shapes are displayed with the same scale, allowing for a consistent visual comparison between them.
- Finally, `clear` and `clc` are used to clear the workspace and the command window at the end of the script.

Overall, this MATLAB script transforms experimentally measured static data into a discrete 6-DOF model of the cantilever plate, from which natural frequencies, mode shapes and 3D visualisations are obtained. These theoretical results can then be compared directly with finite element predictions and with the experimental modal patterns observed using sand on the vibrating plate.

Chapter 5

Procedure SolidWorks

For the numerical analysis of the cantilever plate, the design and simulation software SOLIDWORKS was used. The process began with the creation of the geometric model of the plate, following the dimensions provided in the assignment brief. In this phase, the rectangular surface was defined and the bolt holes were incorporated, each with a diameter of 3 mm, in order to faithfully reproduce the physical configuration studied in the laboratory.

Once the main geometry was established, the bolts were added by importing them from the SolidWorks component library. Each bolt was positioned precisely in its corresponding hole, ensuring that the mass distribution and the boundary conditions accurately represented the real assembly. In this way, the numerical simulation reflects the expected structural behaviour consistently. The following figures show the fully assembled model before proceeding with the simulations.

Once the geometry of the plate was completed, the next step consisted of running the numerical simulations by assigning the appropriate material properties. In SOLIDWORKS, these parameters are defined through the material specification panels, where all mechan-

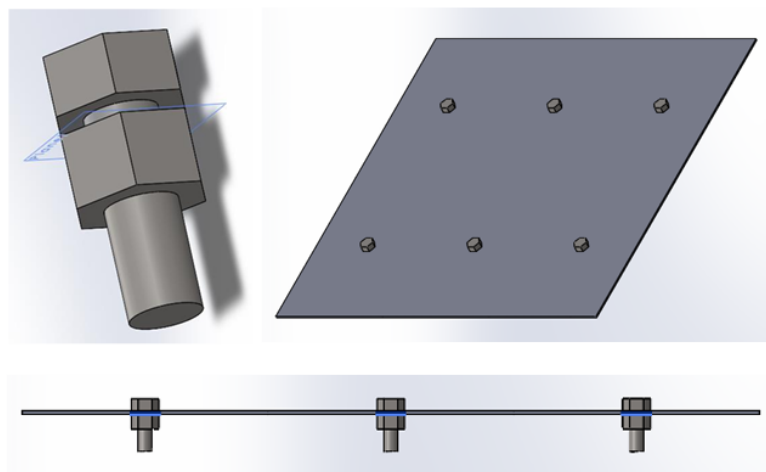


Figure 5.1: Fully defined geometry displayed before initiating the simulation workflow

Properties Tables & Curves Appearance CrossHatch Custom Application Data Favi

Material properties
Materials in the default library can not be edited. You must first copy the material to a custom library to edit it.

Model Type: Linear Elastic Isotropic ☐ Save model type in library

Units: SI - N/m² (Pa)

Category: Plastic

Name: 1060 Alloy

Default failure criterion: Max von Mises Stress

Description:

Source:

Sustainability: 1060 Alloy in SOLIDWORKS Materials : Aluminium Alloys

Property	Value	Units
Elastic Modulus	6.9e+10	N/m ²
Poisson's Ratio	0.33	N/A
Shear Modulus	2.7e+10	N/m ²
Mass Density	2770	kg/m ³
Tensile Strength	68935600	N/m ²
Compressive Strength		N/m ²
Yield Strength	27574200	N/m ²
Thermal Expansion Coefficient	2.4e-05	/K
Thermal Conductivity	200	W/(m-K)

Figure 5.2: Specification of Aluminium 1060 as the cantilever plate material in Solid-Works, with density set to $2770 \text{ kg}\cdot\text{m}^{-3}$

ical characteristics relevant to the structural behaviour of the model are introduced.

For this study, the cantilever plate was modelled using an Aluminium 1060 alloy. Although this material typically exhibits a nominal density of approximately $2705 \text{ kg}\cdot\text{m}^{-3}$, it was necessary to adjust this value to $2770 \text{ kg}\cdot\text{m}^{-3}$ to remain consistent with the information provided in the assignment brief. This modification ensures that the mass of the digital model matches the mass used in the theoretical calculations and in the experimental setup, thereby enabling a reliable comparison between numerical, analytical, and physical results.

Accurate definition of the material properties is essential, as both the mass distribution and structural stiffness depend directly on these parameters. Any variation in their values may influence the resulting natural frequencies and mode shapes obtained from the simulation. The following figures show the final material configuration applied before initiating the simulation process.

To replicate the physical setup used in the experiment, the plate was fixed along one of its edges, leaving the remaining surface free to vibrate. After applying this boundary condition, a fine mesh was generated across the plate to ensure that the simulation captured the structural response with sufficient detail. Both the mesh and the fixed support can be seen in the figure below.

Once the model was prepared, the simulation parameters were defined. In this study,

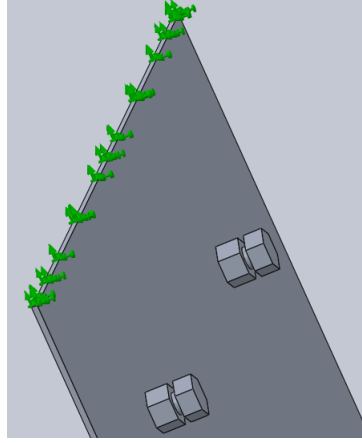


Figure 5.3: boundary conditions

ten vibration modes were requested, corresponding to the first ten natural frequencies of the system. SolidWorks then carried out the modal analysis and produced ten results, each showing the deformation pattern of the plate and the frequency at which it occurs.

These modes and their associated displacement fields will be presented and discussed in the Results section of this report.

Numerical Compliance Matrix Analysis

To obtain the compliance matrix using SOLIDWORKS, a series of static simulations was carried out. In each simulation, one bolt location was loaded vertically while the remaining bolts were left unloaded, reproducing the same procedure used in the laboratory. For every load case, the software provided the vertical displacement at the six measurement stations.

Measuring Station ↓	Loading Station →					
	1	2	3	4	5	6
1	1.5 kg	1.5 kg	1 kg	1 kg	1 kg	1 kg
2	1.5 kg	1.5 kg	1 kg	1 kg	1 kg	1 kg
3	1 kg	1 kg	1 kg	1 kg	1 kg	1 kg
4	1 kg	1 kg	1 kg	1 kg	1 kg	1 kg
5	1 kg	1 kg	1 kg	1 kg	0.5 kg	0.5 kg
6	1 kg	1 kg	1 kg	1 kg	0.5 kg	0.5 kg

Figure 5.4: Loading scheme used in the static simulations.

Using the simulated displacements $\delta_{i,j}$ and the corresponding applied force f_j , each coefficient of the numerical compliance matrix was obtained through the classical relation:

$$h_{i,j} = \frac{\delta_{i,j}}{f_j}.$$

This approach mirrors the experimental method but benefits from the additional detail present in the finite element model. Because SOLIDWORKS accounts for the actual plate

geometry, the bolt holes and their influence on local stiffness, the resulting compliance matrix captures the spatial variation in flexibility more realistically than the simplified 6-DOF analytical model.

A comparison between the numerical and theoretical matrices shows the same overall trends: regions farther from the clamped edge are more flexible, while locations close to the fixation are noticeably stiffer. Small differences appear due to the assumptions in the theoretical model, such as uniform material properties and perfectly rigid boundaries, and from discretisation effects in the FE simulation.

Overall, the compliance matrix obtained from SOLIDWORKS provides a clearer and more accurate representation of the plate's static behaviour and serves as a solid reference when interpreting the differences between numerical predictions and experimental modal results.

Numerical Compliance Matrix (in m/N)					
9,81312E-06	2,73191E-06	3,1723E-05	1,7615E-05	4,95209E-05	3,051E-05
2,73191E-06	1,01733E-05	1,7604E-05	3,1723E-05	3,60245E-05	5,4597E-05
2,86442E-05	1,58818E-05	0,00018287	0,0001159	0,00034577	0,00024117
1,58919E-05	2,8634E-05	0,0001159	0,00018277	0,000256065	0,00040581
4,62487E-05	3,21814E-05	0,00034251	0,00025352	0,000789399	0,00055982
3,22018E-05	4,62283E-05	0,00025362	0,00034241	0,000578573	0,00064924

Figure 5.5: Displacements extracted from SolidWorks used to build the numerical compliance matrix.

Chapter 6

Results

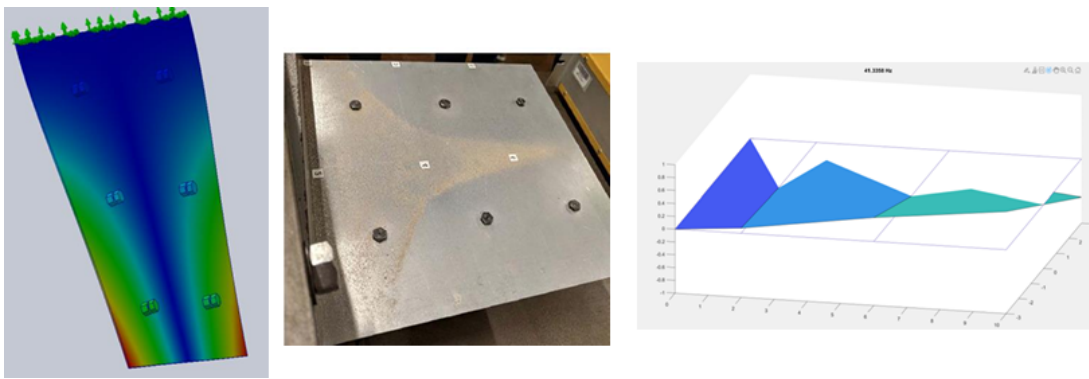


Figure 6.1: Comparison of the second resonance and mode shape: SolidWorks (left) at 34.836 Hz, experiment (centre) at 37.19 Hz, MATLAB (right) at 40.3358 Hz.

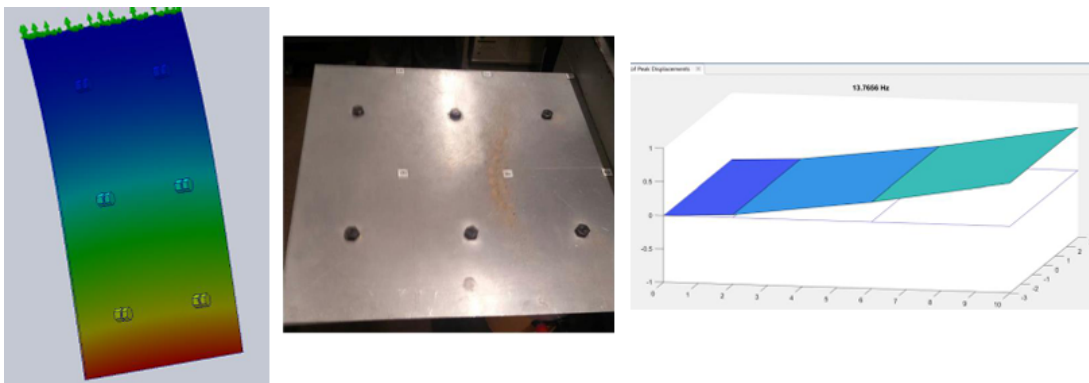


Figure 6.2: Comparison of the first resonance and mode shape: SolidWorks (left) at 14.24 Hz, experiment (centre) at 13.8 Hz, MATLAB (right) at 13.7656 Hz.

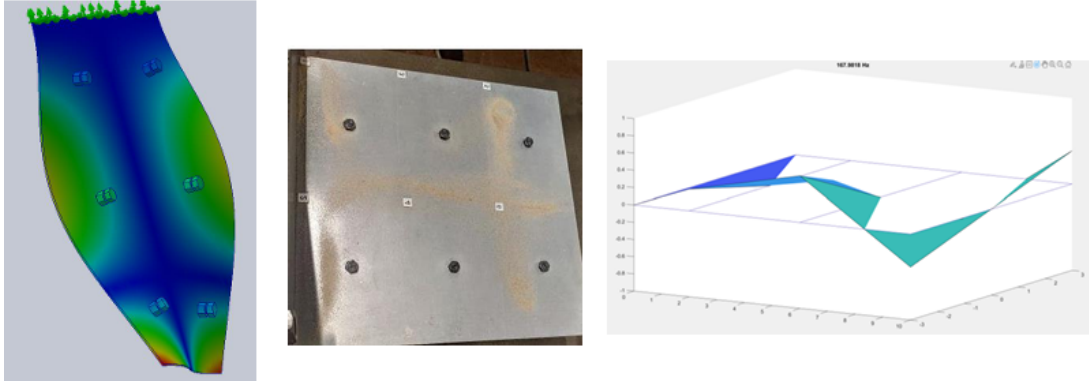


Figure 6.3: Comparison of the fifth resonance and mode shape: SolidWorks (left) at 128.7 Hz, experiment (centre) at 123.6 Hz, MATLAB (right) at 167.981 Hz.

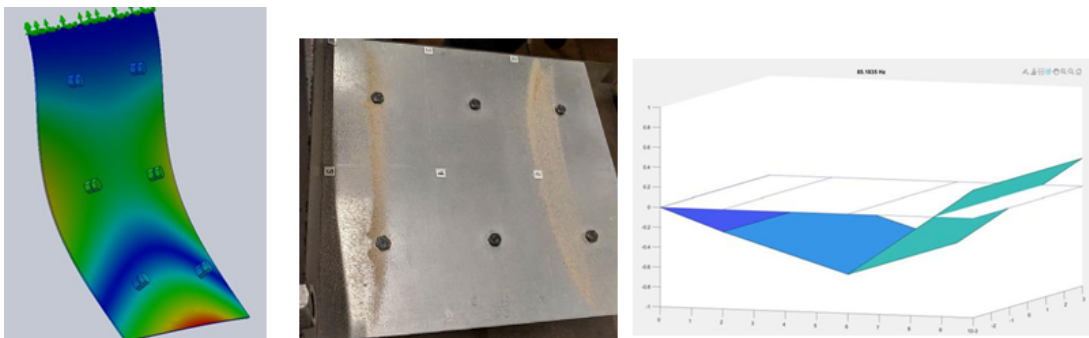


Figure 6.4: Comparison of a higher-order resonance: SolidWorks (left) at 88.209 Hz, experiment (centre) at 84.78 Hz, MATLAB (right) at 85.1835 Hz.

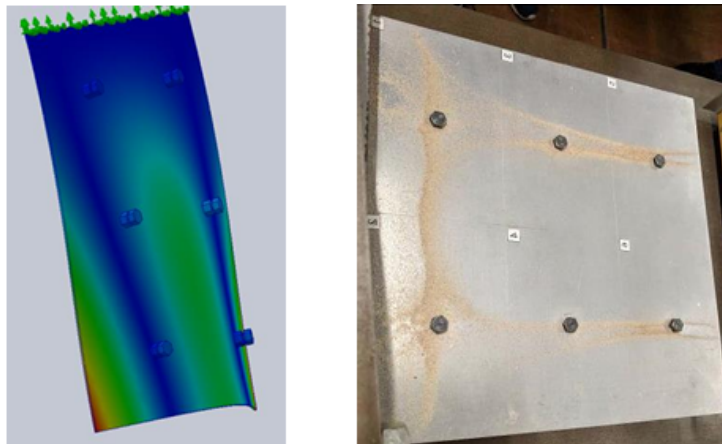


Figure 6.5: Resonance identified at 113.81 Hz in SolidWorks (left) and at 109.0 Hz experimentally (centre); no MATLAB plot is available for this frequency.

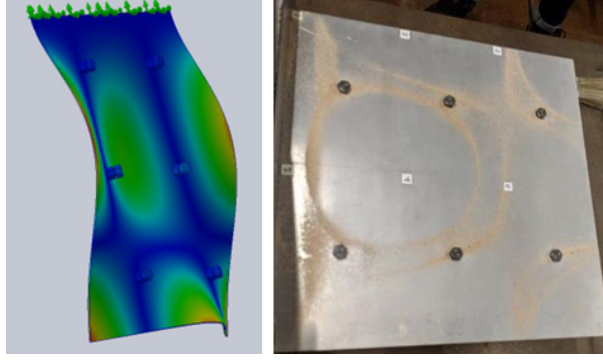


Figure 6.6: Resonance identified at 231.62 Hz in SolidWorks (left) and at 205.0 Hz experimentally (centre); no MATLAB plot is available for this frequency.

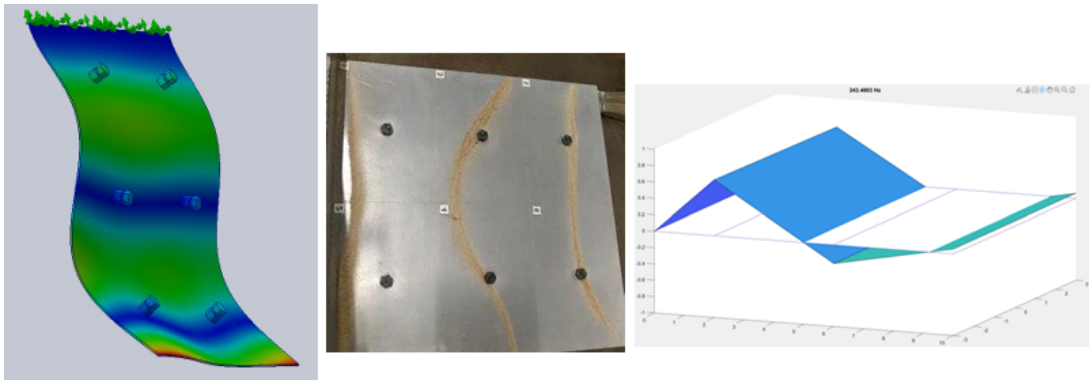


Figure 6.7: Comparison of another resonance: SolidWorks (left) at 259.54 Hz, experiment (centre) at 236.2 Hz, MATLAB (right) at 243.4803 Hz.

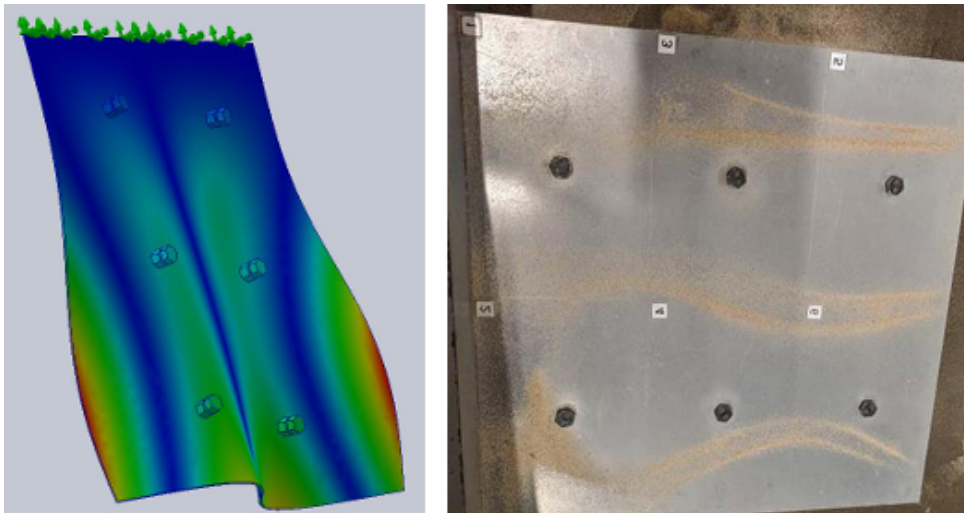


Figure 6.8: Resonance identified at 264.62 Hz in SolidWorks (left) and at 245.0 Hz experimentally (centre); no MATLAB plot is available for this frequency.

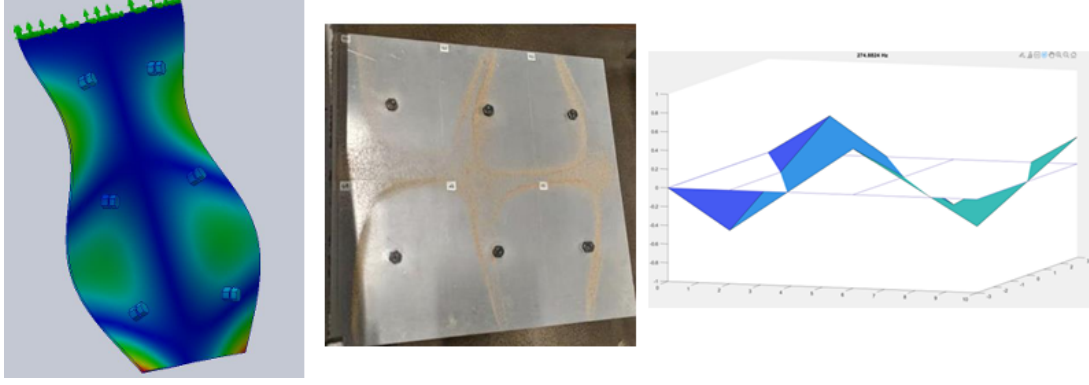


Figure 6.9: Comparison of a higher resonance: SolidWorks (left) at 302.86 Hz, experiment (centre) at 281.90 Hz, MATLAB (right) at 274.8824Hz.

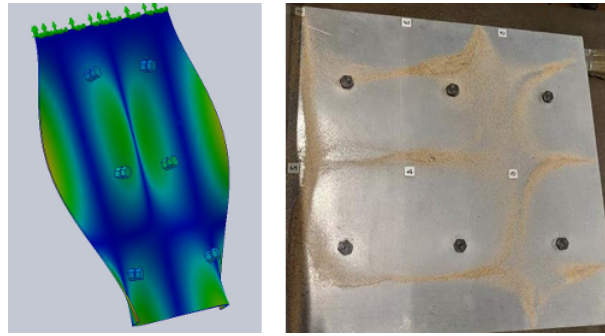


Figure 6.10: Resonance identified at 387.85 Hz in SolidWorks (left) and at 337.10 Hz experimentally (centre); no MATLAB plot is available for this frequency.

Discussion

A comparative analysis of the three approaches reveals not only differences in the results but, more importantly, the underlying reasons for these discrepancies. The experimental method, although subject to noise and handling variability, represents the real physical system in its full complexity. The redistribution of sand across the plate surface highlights the nodal regions with great clarity, and these experimentally observed patterns often expose behaviours that simplified numerical models fail to capture. For instance, subtle asymmetries in the clamping condition, small imperfections in the plate's geometry, and material inhomogeneities naturally alter the mode shapes and resonance frequencies. These effects are very difficult to replicate exactly in a theoretical or numerical model.

The SolidWorks simulations reproduce the experimental mode shapes with high accuracy because the finite element method inherently accounts for local variations in stiffness and mass, as well as the geometric discontinuities introduced by the bolt holes. However, FEM is still sensitive to the assumptions embedded in the model. The clamped boundary condition, for example, is treated as perfectly rigid, whereas the physical clamp exhibits some compliance, which reduces the experimental frequencies. Likewise, mesh density

significantly affects the accuracy of the higher-order modes: insufficient refinement causes numerical stiffening, which artificially increases the predicted natural frequencies. This explains why SolidWorks maintains excellent agreement for the first modes but gradually diverges as the deformation patterns become more intricate and require finer spatial resolution.

In contrast, the MATLAB approach is fundamentally limited by its formulation. The 6-degree-of-freedom model is a drastic simplification of a continuous elastic medium. By condensing the entire plate into six lumped masses connected by equivalent springs, the method implicitly assumes that the deformation between measurement stations is smooth and can be described with low-order curvature. This is sufficient for the first modes, where bending is dominant and varies gradually along the plate. However, higher modes exhibit multiple inflection points and localised curvature changes. A lumped model with only six nodes cannot reproduce these geometric features and instead forces the structure to vibrate in shapes that are mathematically admissible within the coarse discretisation, but physically unrealistic.

Furthermore, the stiffness matrix in MATLAB is derived from the experimental compliance matrix, which itself contains measurement noise. When inverted numerically, even small errors in the compliance matrix lead to significant distortions in the stiffness representation. These distortions have a minimal effect on the lower eigenvalues but amplify dramatically for higher modes, producing large deviations in the predicted frequencies. Therefore, MATLAB does not fail simply because the mathematical model is simplified, but also because the method is highly sensitive to noise—far more than FEM, which distributes errors across thousands of degrees of freedom.

Taken together, these findings show that each method highlights different aspects of the structure’s behaviour: the experiment captures the full physical reality (but with noise), FEM captures the geometry and material response with high fidelity (but is sensitive to modelling idealisation), and MATLAB captures only the global dynamic trends (but breaks down when local curvature becomes dominant).

Summary of mode shape frequencies and relative error

The numerical and experimental comparisons are further clarified by a summary table that compiles, for each resonance mode, the corresponding natural frequency obtained from the three approaches: experiment, MATLAB and SolidWorks. Using a spreadsheet, the relative error between each numerical prediction and the experimental value was calculated. This quantitative analysis complements the visual comparison of mode shapes, showing not only that the qualitative deformation patterns agree, but also how close the predicted frequencies are to the measured ones for each method and each mode.

Mode Shape	Resonance Frequency Values			Difference between values generated by ...	
	MATLAB Frequency (Hz)	SolidWorks Frequency (Hz)	Experiment Frequency (Hz)	MATLAB and SolidWorks	SolidWorks and Experiment
1st Mode	13,77	14,24	13,8	0,47	0,44
2nd Mode	41,34	34,836	37,19	6,504	2,354
3rd Mode	84,78	88,209	84,78	3,429	3,429
4th Mode	No result	113,81	109	No result	4,81
5th Mode		128,7	123,6	128,7	5,1
6th Mode	No result	231,62	205	No result	26,62
7th Mode	243,48	259,54	236,2	16,06	23,34
8th Mode	No result	264,62	245	No result	19,62
9th Mode	274,9	302,86	281,9	27,96	20,96
10th Mode	No result	387,85	337,1	No result	50,75

Figure 6.11: Consolidated table of outcomes and evaluation of discrepancies across the methods.

Frequency Comparison and Error Analysis

The comparison of the resonance frequencies obtained from MATLAB, SolidWorks and the experimental measurements highlights clear differences in accuracy across the modal range. MATLAB performs very well in the lower modes, with errors below 1% in the first and third modes, showing that the simplified 6-DOF model is capable of capturing the global bending behaviour of the plate. However, the error increases noticeably for higher modes: the second mode differs by 11.2% and the fifth by 4.13%. These deviations indicate that the reduced model lacks the spatial resolution needed to represent the more complex shapes that appear at higher frequencies. In addition, because the stiffness matrix is obtained by inverting the experimentally measured flexibility matrix, any measurement noise is amplified, which further contributes to the loss of accuracy in the upper modes.

SolidWorks shows a more stable performance across most of the modal range. In the first five modes, the error stays between 3% and 6%, demonstrating that the finite element model captures the stiffness and mass distribution of the structure with reasonable fidelity. Beyond this range, however, the error grows significantly: the seventh and ninth modes differ from the experiment by nearly 10%, and the tenth shows a deviation of 15.05%. These increasing errors reflect the sensitivity of higher-order modes to mesh refinement and boundary modelling. Small imperfections in the real clamp, or insufficient mesh density near regions of high curvature, have a much stronger influence on the higher modes than on the lower ones.

Overall, MATLAB provides accurate predictions for the fundamental modes but becomes unreliable at higher frequencies. SolidWorks offers better agreement over a wider modal range, though it also loses accuracy in the most complex modes. The experimental results therefore remain essential as a reference for evaluating where each numerical model begins to deviate from the real dynamic behaviour of the plate.

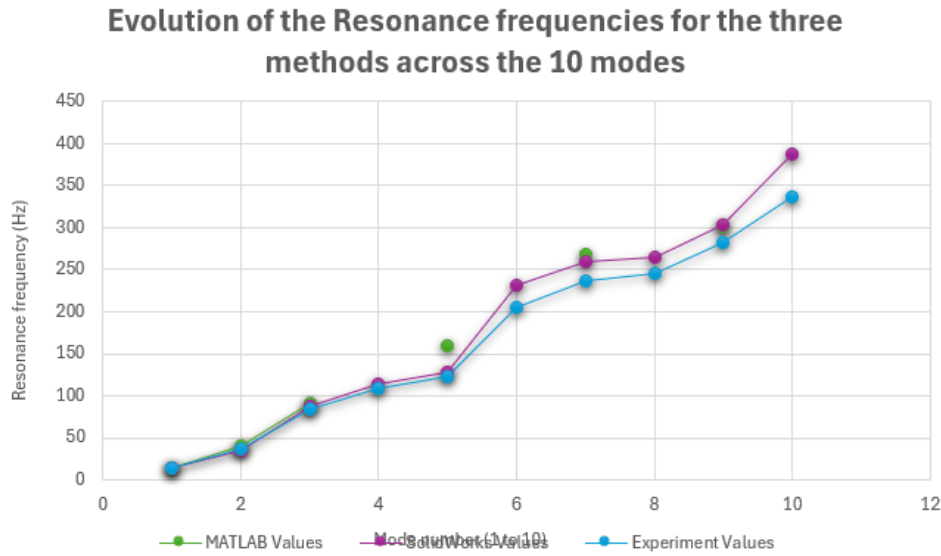


Figure 6.12: Plot showing the progression of resonance frequencies across the ten modes for all three methods.

Analysis of Divergence Across Higher Modes

The graph illustrates how the discrepancy between the three methods becomes progressively more pronounced as the vibration modes increase. Higher-order modes exhibit complex deformation patterns with multiple nodal and anti-nodal regions, making them particularly sensitive to modelling assumptions, boundary conditions and mesh refinement. As a result, even minor inaccuracies can lead to significant deviations in the predicted natural frequencies.

This behaviour explains the strong divergence observed in the MATLAB results from the fifth mode onward. The simplified 6-DOF lumped model cannot capture the intricate dynamic interactions that characterise higher-frequency responses. SolidWorks performs considerably better in the lower modes, but its accuracy also deteriorates as the modal shapes become more complex, revealing the limits of a moderately refined finite element mesh and idealised boundary conditions.

The trend in the graph confirms this interpretation. Beyond the fifth mode, the difference between MATLAB and SolidWorks decreases—not because MATLAB becomes more accurate, but because SolidWorks begins to struggle with resolving the increasingly complex vibration shapes. At the same time, the discrepancy between SolidWorks and the experimental measurements continues to grow, indicating that even the FEM model no longer fully represents the real physical behaviour.

For this reason, experimental data become essential when analysing higher-order modes, as relying solely on simplified or moderately refined numerical models may lead to substantial errors. More advanced simulation tools, such as ANSYS with adaptive

meshing and detailed boundary modelling, would be required to obtain a more reliable representation of these complex modal responses.

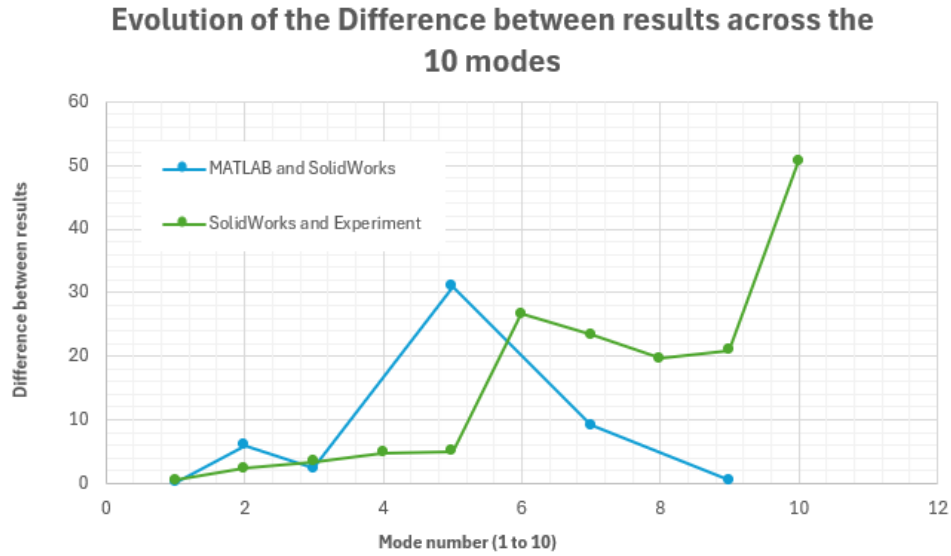


Figure 6.13: Plot showing how the discrepancies between the different methods vary across the ten modes.

Comparative Analysis of the Three Approaches

A critical comparison of the experimental measurements, the SolidWorks finite element simulations, and the reduced MATLAB model reveals that the discrepancies between these methods stem not only from differences in modelling philosophy but also from fundamental assumptions about the physics of the problem. Although all three approaches aim to predict the same modal characteristics, they operate within very different levels of abstraction, and this directly influences their accuracy and reliability across the full range of vibration modes.

Experimental Approach

The experimental procedure provides the most faithful representation of the plate's behaviour because it incorporates every physical effect inherently present in the real system. These include: material imperfections, minor variations in thickness, subtle compliance in the clamping device, and natural damping. Such factors shift the natural frequencies and alter the nodal patterns in ways that are almost impossible to reproduce numerically without highly detailed modelling. Therefore, the experiment becomes the reference against which all other methods must be evaluated.

However, the strength of the experimental method is also its main weakness: it is highly sensitive to disturbances. Noise in the laboratory environment, slight inconsis-

tencies in the shaker input, and the redistribution of sand—which is not perfectly controlled—introduce variability. As a result, the experimentally measured frequencies fluctuate within a range, rather than converging to an exact value. This makes the experiment reliable in a physical sense, but less repeatable than numerical methods.

SolidWorks Finite Element Simulations

The SolidWorks simulations demonstrate the strongest overall performance because the finite element method distributes stiffness and mass continuously across the structure. This allows the model to capture local curvature, geometric discontinuities, and boundary effects more accurately than analytical lumped models.

Nonetheless, FEM predictions remain influenced by two critical factors:

- **Boundary idealisation:** The clamp is assumed perfectly rigid, whereas the real clamping device exhibits finite stiffness. This systematically increases the computed natural frequencies, since a fully rigid support artificially stiffens the system.
- **Mesh density:** Higher-order modes involve rapid spatial variations in displacement. If the mesh is not sufficiently refined, the simulation becomes numerically stiff, producing modal frequencies that drift progressively away from the experimental values.

Despite these limitations, SolidWorks reproduces the nodal patterns with remarkable fidelity. Even intricate modal shapes—such as those involving multiple internal nodal surfaces—are accurately captured, demonstrating that FEM is well-suited for this class of problems as long as the mesh is properly resolved.

MATLAB Reduced-Order Model

The MATLAB model offers a completely different perspective. By discretising the plate into only six lumped degrees of freedom, it inherently assumes that the deformation between nodes is smooth and that each segment behaves like a simple beam element. This approximation is valid for the lower modes, which are governed primarily by global bending behaviour. Consequently, MATLAB predicts the first few natural frequencies with reasonably small error.

However, as soon as the mode number increases, the limitations become unavoidable:

- **Loss of spatial resolution:** Higher modes introduce multiple inflection points that cannot be represented by a 6-DOF discretisation. The model forces the plate into unrealistic deformation shapes, causing large frequency errors.

- **Sensitivity to measurement noise:** The stiffness matrix used in MATLAB is obtained by numerically inverting the experimental compliance matrix. Because matrix inversion amplifies noise by several orders of magnitude, small uncertainties in the displacement measurements lead to substantial distortions in the computed stiffness.

The combined effect of coarse discretisation and noise amplification explains why the MATLAB predictions diverge so sharply in higher modes, both in frequency and in mode shape topology.

Overall Interpretation

Collectively, these findings reveal that no single method is universally superior; instead, each provides reliable information only within the range of assumptions under which it was developed. The experiment captures the full physical complexity but suffers from noise; FEM provides a high-fidelity numerical approximation but relies on mesh quality and idealised boundaries; and MATLAB delivers fast predictions but only for the global behaviour of the first modes. When interpreted together, however, the three methods provide a coherent and multi-layered understanding of the dynamic behaviour of the cantilever plate.

Chapter 7

Conclusions

This study shows that the three approaches used—MATLAB, SolidWorks and the experimental measurements—offer complementary insights into the dynamic behaviour of the cantilever plate, but with different levels of accuracy.

MATLAB performs well for the fundamental modes, with errors below 1% in the first and third modes. This confirms that the simplified 6-DOF model can capture the global bending behaviour of the plate. However, the error increases significantly in higher modes (11.2% in the second mode and over 4% in the fifth), mainly because the coarse discretisation cannot represent complex deformation shapes and because the stiffness matrix, obtained from experimental data, amplifies measurement noise during inversion. Thus, MATLAB remains useful for quick preliminary analysis but is not reliable for higher-order dynamics.

SolidWorks provides a much closer match to the experiment across most of the modal range. In the first five modes, the error stays within 3–6%, indicating a good representation of the plate’s geometry and stiffness distribution. The accuracy decreases in the higher modes, where errors reach 10–15%, due to the sensitivity of high-frequency vibration shapes to mesh density and boundary idealisation. Despite these limitations, SolidWorks offers the most consistent numerical predictions overall.

The experiment, although affected by human error and practical limitations, represents the true behaviour of the structure. Small variations in clamping, imperfections in sand distribution and slight frequency-reading inaccuracies are unavoidable, yet they reveal physical effects that numerical models idealise.

In summary, MATLAB is effective for understanding fundamental behaviour, SolidWorks provides the most accurate numerical results across the modal range, and the experiment remains essential for validating both. Together, these methods give a clear and balanced understanding of the vibrational characteristics of the cantilever plate.

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